

MAHESH LOYA

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SUMMARY

Full-stack and AI engineer in the final year of a B.E. in Information Technology, passionate about building AI-native software with real-world impact. Comfortable owning products end-to-end — from architecture and data modelling to user experience, deployment, and iteration after launch.

EDUCATION

Pune Institute of Computer Technology <i>B.E. in Information Technology</i> <i>Relevant coursework:</i> Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP Marathwada High School & Junior College — HSC (Class XII) Oasis's English School — SSC (Class X)	Nov 2023 – May 2027 <i>CGPA: 8.66 / 10</i> 76.67% · 2023 100% · 2021
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EXPERIENCE

Software Developer (Freelance) — Vyavsay Assist <i>Independent engagement: Girija Motors</i> - On-site discovery at the dealership showed high-intent leads were being lost to after-hours WhatsApp enquiries nobody answered. Built an AI sales copilot that handles text, Hindi/Hinglish voice notes and customer-sent car photos through one conversational pipeline. - Architected the backend in TypeScript and Fastify on PostgreSQL, integrating GPT-4o Vision for make-and-model detection, Whisper for speech-to-text, and tool-calling for inventory search, appointment booking and human escalation. - Implemented RAG over live dealer inventory on pgvector (1536-dim, HNSW index), fusing semantic search with SQL filters so numeric constraints (price, year) are enforced exactly rather than approximated. - Built the dealer-facing CRM dashboard in React/Vite, and deployed on AWS EC2 with Docker, Nginx and rate limiting; in daily use at the dealership, shipped on a weekly feedback cycle with the sales team.	Feb 2026 – Present <i>Pune, India</i>
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PROJECTS

Blood Donation Drive Management System (<i>live</i>) <i>React, Node.js, Express, MongoDB, Passport.js</i> - PICT's blood donation drives ran on paper — donors queued to fill forms, organisers tracked turnout by hand. Built a full-stack platform that digitised the drive end to end: online sign-up, automated eligibility screening and organiser dashboards. - Developed REST APIs with session-based authentication — Passport.js local strategy with salted password hashing, sessions persisted in MongoDB — and server-side eligibility validation (age and weight); modelled the donor schema around a status workflow (pending → approved → completed/rejected) with timestamp tracking per transition. - Built MongoDB aggregation pipelines (\$match + \$group) for turnout and blood-group reporting. Deployed on the official Pictoreal domain — 231+ donor registrations processed, cutting organiser coordination from hours of manual tracking to a single dashboard.	
Voice AI Receptionist <i>Python, FastAPI, PostgreSQL, Redis, Docker</i> - Small businesses lose inbound enquiries to calls nobody answers, and a missed call leaves no record. Built the real-time pipeline for a multi-tenant voice receptionist qualifying callers in Hindi, Marathi and English — a VAD → STT → LLM → TTS streaming stack with a FastAPI media bridge over WebSockets. - Designed tenant isolation end to end: a 7-table tenant-scoped schema (SQLAlchemy, Alembic), Redis call state under a mandatory TTL, and a DID→tenant cache whose bounded TTL is a correctness control — a reassigned number can never keep routing to its previous tenant. - Solved barge-in (caller interrupts the bot) by identifying that halting generation still leaves the playout buffer playing, fixing it with an explicit clear control frame; validated via 12 recorded fixtures in a mock-carrier harness needing no telephony account.	

TECHNICAL SKILLS

Languages	C++, JavaScript, Python, SQL
Frontend	React, HTML/CSS
Backend	FastAPI, Node.js, Fastify, Express, REST, WebSockets, SQLAlchemy, Alembic
Databases	PostgreSQL, MongoDB
AI / LLM	GPT-4o, Claude, Gemini; RAG, vector search, embeddings, prompt & context engineering, agentic tool use
Cloud	AWS, Docker, Nginx, Linux, Git, CI/CD

ACHIEVEMENTS & CERTIFICATIONS

- Winner, MVPMM Hackathon 2026** — 1st prize of **Rs. 1,00,000** for building Vyavsay Assist, an AI WhatsApp sales copilot for Indian SMBs.
- Best Event Coordinator Award**, Pictofest 2025. **2-Star** on CodeChef; **100+ DSA problems** solved on LeetCode.
- Certs:** DeepLearning.AI — Vector Databases | Claude 101 | Full Stack Web Dev | Mastering DSA.

POSITIONS OF RESPONSIBILITY

Overall Event Coordinator, Pictofest, PICT Pune Directed 500+ volunteers across 20+ events, 2,000+ attendees from 22+ colleges; secured Rs. 1,00,000 in sponsorships.	Oct 2024 – Feb 2025
Photography Team Head, Pictoreal, PICT Pune — <i>Lead media team; mentor juniors</i>	Jul 2025 – Present